ferrules. We are applying these flues without prossering them or beading them. The hole is drilled to suit the flue without the copper ferrule, which automatically increases the width of the bridge. Then the flue is belled over to 45 degrees. We weld around the belled-over end of the flue.

As I say, we are just turning out these two engines, but we feel that this method of application will be beneficial. Years ago we welded flues with hot water in the boiler and we have started that practice again. We believe it will tend to control the heat penetration and eliminate the starting of bridge cracking and probably help out on the fire cracking of flues. At the next meeting we will be able to report any benefits derived.

H. H. SERVICE: In regard to coppers projecting from heads on flues, the arc must penetrate the metal and, if the copper is there, it will draw, whether the shielded arc or any other is used. The coppers must be kept back. Reversed polarity with the shielded-arc process, of course penetrates slightly deeper. That is one of the principal reasons why you must fit plates closer.

In regard to Mr. Pool's remarks, of welding flues in two engines with the ends of flues flared out, I might say that the Santa Fe did some of this work from 1908 to 1913 on what was known as Jacob's superheaters. However, the work was done by the oxy-acetylene process and all welding was done downhanded. Nevertheless, the work did not stand up, which of course, could not be termed as a black eye against the oxy-acetylene process of today. Nevertheless, it did not work out satisfactorily, and I might add that later on this superheater arrangement was termed "Jacob's refrigerator."

By all mean keep hot water in the boiler when welding flues. About 90 degrees F. is our practice.

J. A. DOARNBERGER (Norfolk & Western): It is a well-known fact that the first completely electrically welded locomotive firebox ever built in this country was the old 778 in July, 1914, at the Roanoke shops of the Norfolk & Western Railroad. That engine is still running but we must go further.

When the Lukens Steel Company put in their 212-inch mill, they only guaranteed to roll 196-inch width sheets. About five years ago we built twenty Mallets carrying 300 pounds pressure with fireboxes for which we wanted sheets 214 inches long. The best we could get was 196, so that left us slack about 18 inches. We solved this problem by welding a triangular piece in the sheet to give us the required size for forming the firebox.

We are now starting on a group of engines of still greater dimensions. For this order we require a sheet 22 feet 4½ inches wide, to be exact, and still the best we can get is 196 inches. This leaves 72 inches short on the width to build this firebox with a 96-inch combustion chamber. Flues for this boiler are 24 feet 2¾ inches in length. The locomotive will operate at a working pressure of 300 pounds per square inch.

What we will have to do is obtain the largest sheets possible and weld them together. That is our only salvation. The work must of course be done with the best welding rod and technique.

Instead of building a firebox with four sheets—two side sheets, crown sheet and bottom of combustion chamber—we have adopted the one-piece sheet. Our only means for building such a firebox is electric welding. It has come to stay.

Wonderful progress has been made. I have always predicted and the time is coming when we will be allowed to weld any part of the stayed portion of the boiler. We have kept off the barrel sheets, but believe me, barrel sheets will be welded just as other pressure vessels.

E. E. OWENS (Union Pacific): I have five sets of 3½-inch flues applied without any coppers, and I did not prosser them in the sheet. I beaded them with an ordinary beading tool and welded them on application. One set has now made about 50-000 miles, and there has not been a leaky flue up to date.

We have districts where we have cracks develop from the heads lengthwise into the flue. When the flue vibrates and loosens in the sheet throwing the stress all on the weld, scale forms in the space. This becomes overheated and causes fire cracks. There are no more fire cracks there than there are in a seam. Flues because they are welded cannot be forgotten. When inspection shows them to be loose tighten them in the hole and you will overcome fire cracking. In some districts you will have to do it oftener than in others. I have districts where we have completely treated water, in California, for instance, where I can run the flues for a year without expanding. In Oregon I have to expand them every thirty days.

When I went up there five years ago, we were getting 45,000 to 50,000 miles on flues. We were taking them out because they were fire cracked. Today we require 100,000 miles from flues and we are making it.

I have one bad condition over on the Union Pacific in our coal burners. That is with cinder cutting. If it were not for welding, I would not get 50,000 miles out of the flues in these locomotives. These are Type E superheated boilers, with 262 flues in the set. In about 16 or 18 months the beads are cut off. The only way I can save them is to catch them before they get too thin and build them up with welding. The reason I did away with coppers is that I did not want too much material at the ends. I wanted a straight flue so that the prosser mark would not catch the cinders.

PRESIDENT FOGERTY: Has not your road adopted the idea of lowering the arch tube at the back end and carrying a longer arch to help eliminate cinder cutting?

E. E. OWENS: We have one engine, a 9,000 type, that now has 90,000 miles. We lowered the arch in the door sheet about 10¾ inches. When cinder cutting first started we carried seven courses in our brick arches. The cinder cutting developed so fast we lowered that to six courses, which slowed it up some. On the engine in which we lowered the arch, I have 90,000 miles. I will make 150,000 miles more easily than 90,000 miles with other engines on which the arch tubes have not been lowered. I have authority now to lower them in 10 more engines.

I am also going to extend it to our Mountain type, where we will lower tube 8½ inches. In these five engines that I have equipped without any coppers, I increased the thickness of the flue sheet 1/16 inch in order to give more bearing and withstand cinder cutting longer. They have not been in service long enough to determine how long they will run.

PRESIDENT FOGERTY: I also believe you have had some trouble on arch tubes leaking at the door sheet and you have helped considerably by building up on the sheet ⅜ inch with electric welding.

E. E. OWENS: About ten years ago the Southern Pacific was doing our work at Ogden. We were running our engines to Evanston. I was sent out there because Ogden could not turn out power fast enough. I went to see what was the matter, and in every case found it would be an arch tube leaking in the door sheet. So I went back to Evanston and started pulling out arch tubes. I found that the tubes were being almost cut off by the sheet. I reinforced the hole with ¼-inch boss around it; reamed it out and put the tube back. This eliminated our trouble in the door sheet.

Sometimes one became loose in the throat sheet, so I built them up on both ends. That is my present practice. I make a ⅝-inch bearing at either end. I have also done away with the copper around the arch tube, putting arch tubes in close-fitting holes, reinforced by the ⅝-inch bearing. Then the tubes are rolled metal to metal. They are tight enough so that any expansion and contraction will go in the curvature of the tube and take care of it.

J. W. KELLY (National Tube Company): In my travels I recently saw an interesting development along the lines of Mr. Owen's practice. The boiler maker doing the work, punches a small hole in the sheet, and with a small die flanges the door sheet and the flue sheet towards the fire. He does not do any welding, but just presses the hole out like the old flue hole in the flange fire.

O. H. KURLFINKE (Southern Pacific): Mr. Service mentioned the fact that the practice on the Southern Pacific is to set tubes without copper ferrules. He is right. We experimented for years endeavoring to get a tube, especially the Type A superheater tube, to stay tight. The oxy-acetylene process years ago would not work. Later on when electric arc welding became popular, it was more successful. Still we had trouble. Tubes working in the boiler coupled with expansion actually pushed out through the back tube sheets and broke the welds.

It brought us to the point where we had to prosser all tubes. At first, we could not get a prosser or a man or an air hammer strong enough to expand the tubes. The practice was never successful until one of our general foremen conceived the idea of having the tube prossered hot in a machine, a safe end about 8 inches long. It makes a very good prosser. It is very high,—I would say ¼ inch, with the face of it touching the inside of the back tube sheet practically square with the sheet, with a very small fillet of about 1/16 inch taking care of the corner where it enters the tube hole.

The surface of the firebox side of these sheets was countersunk with a bevel of 45 degrees ¼ inch deep or one-half the thickness of the plate. The tube then was belled out, and the end welded to the sheet. There was no projection of the weld on the firebox side. Those sheets are practically flush. We do not permit the men to build the weld up.

Later on when the Type E superheater came into being we did the same thing without any experimentation at all.

I mention this to illustrate the wonderful success that we are getting with these big engines in that the tubes will run their life. We are having absolutely no trouble with leaky superheater tubes. We do follow the practice on the small tubes of running the engines until the tubes give us trouble. We do not weld them in the beginning, but we found out from ex-

(Continued on page 286)

Questions and Answers Pertaining to Boilers

This department is maintained for the purpose of helping those who desire assistance on boiler and plate fabricating problems. Inquiries should bear the name and address of the writer. Anonymous communications will not be considered. The identity of the writer, however, will not be disclosed unless special permission is given to do so.

By George M. Davies

Copper Ferrules in Firetube Boilers

Q.—Being in need of some information about retubing of firetube boilers I am writing to you for same. I am going to help soon in retubing a locomotive crane boiler and since they have talked about it since last spring and with the cold weather coming on they are to retube it soon. I have had experience in boiler making as I worked on it for many years until I had to give it up on account of ill health. About 2 years ago when they were installing a high-pressure boiler here in a plant, one of the men took a contract to retube the boiler I am referring to and he did not put any ferrules in the tube sheet where the holes were not too big, particularly on the outside rows of tubes.

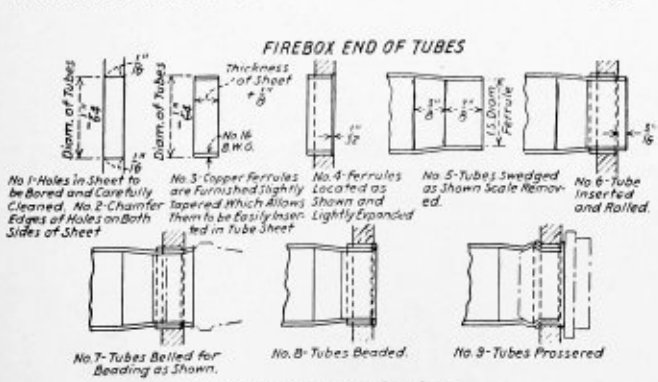
The tubes were put in just as they came from the factory and it seems as though they gave pretty good results. I have never in my experience in retubing boilers failed to use copper ferrules. Our usual procedure was to take a tube, heat and swedge it to a certain size and then try it on the tube sheet to get at an average swedge. I don't seem to be able to see how the tubes would hold any length of time. I have had quite an argument and I would appreciate some advice on this matter. T. R.

A.—The A.S.M.E. Code does not require the use of copper ferrules in firetube boilers (P-250). A firetube boiler shall have the ends of the tubes firmly rolled and beaded, or rolled, beaded and welded around the edge of the bead. When the tubes do not exceed 1½ inches in diameter, the tube sheet may be chamfered or recessed to a depth at least equal to the thickness of the tubes and the tubes rolled into place and welded.

In no case shall the tube end extend more than ⅜ inch beyond the tube sheet. In the case of tubes not exceeding 1½ inches in diameter, they may be expanded by the prosser method in place of rolling.

I-19.—Tube-hole tolerance and use of ferrules. The initial diameter of finished tube holes shall not exceed the nominal diameter of the tubes by an amount greater than shown in the following table.

	Inches
Fire end of firetube boilers	1/32
Opposite end of firetube boilers	1/16
Watertube boilers	1/32



Fitting ferrules in tube sheets

If ferrules are used, proper allowance should be made for the thickness of ferrules. Rough and sharp edges of the tube holes shall be removed as required by the code.

Although copper ferrules are not required, they are still used extensively, especially in locomotive boilers. Copper, as is well known, has a higher range of ductility than any other metal used in mechanical lines and in tenacity it is exceeded only by the better grades of iron. In sheet form it gives a more equal distribution of contraction and expansion stresses than any other metal. Leaky flues are caused mainly by too sudden change of temperature in the firebox.

Under this condition, the copper ferrule comes into its own. While it does not entirely eliminate trouble it stands in the position of a shock absorber or as an equalizer, for then its natural properties are set in motion and a more even contraction is brought about, so that the trouble is not so great as it would be without ferrules.

It is also used as a shim when a tube hole is too large in diameter and out of round from too much expanding and rolling. It would be preferable to use copper ferrules to insure a tight fit between the tube and the tube sheet at all times. The recommended practice for applying tubes in the firebox end of locomotive type boilers is shown in Fig. 1.

The Fitting of Radial Staybolts

Q.—I have been a reader of The Boiler Maker for a great many years, and value your department highly, as I have gotten much valuable information therefrom. Now, I find myself stuck, and turn to you for enlightenment and help. This will probably give you a laugh, but I find myself with the job of renewing a crown sheet, and after 16 years of practical boiler work, including fitting-up, rolling, flanging, flue work and welding, I have never applied a set of radial staybolts, and find I have neither the taps nor the experience to work with.

1. Now, in applying rigid crown stays, say 1⅛ inches diameter by 24 inches long, working pressure 180 pounds, is the taper head bolt or straight bolt preferable?

2. Is it the practice to preserve the continuity of the thread between the upper and lower holes, that is, are the holes both tapped at the same time with the same tap, or are they tapped individually?

3. Please describe the taps and tapping operation in detail for both straight and taper head bolts.

4. Where can these taps be obtained?

5. In your opinion, is the application of copper ferrules in the back flue sheet desirable or necessary when the flues are to be welded to the sheet, in locomotive practice?

I would appreciate it very much if you would give this your prompt attention. J. C. H.

A.—1. The tapered crown stay is preferable to the straight bolt. Tests have demonstrated that a 1¼-inch diameter straight hammered-head crown bolt will support the crown sheet when overheated, under approximately the same conditions as the 1⅛-inch diameter hammered-head crown bolt with a 1½-inch or 2-inch taper in 12 inches. Generally speaking, it can be said that increased diameter of crown bolt at crown-sheet fit gives approximately the same holding power in a heated sheet as in-